

Beyond Failing to Reject: A Bayesian Approach to Evaluating the Andilaye WASH Intervention

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Stage 1: Research Question and Data

Background and Motivation

The Andilaye trial was a parallel cluster-randomized controlled trial conducted in rural and peri-urban Amhara, Ethiopia, testing whether a demand-side water, sanitation, and hygiene (WASH) promotion intervention could improve sustained WASH behavior and mental well-being relative to standard programming (Delea et al., 2019; Freeman et al., 2022). Fifty kebeles across three woredas — Farta, Fogera, and Bahir Dar Zuria — were randomly assigned to intervention or control, with household-level outcomes measured at baseline and approximately two years later at endline.

The original published analysis using Generalized Estimating Equations (GEE) and a subsequent reanalysis using mixed-effects logistic regression both yielded non-significant results for all three mental health outcomes. However, a non-significant p-value cannot distinguish between two substantively different situations: (1) the intervention truly has no meaningful effect, or (2) the study had insufficient power to detect a real effect. This ambiguity is particularly consequential in implementation science, where null results are frequently used to de-prioritize or discontinue programs without any quantification of how confident we actually are that effects are small.

A Bayesian reanalysis resolves this ambiguity in two ways. First, by computing the posterior probability that the treatment effect on the probability scale — expressed as an Average Predictive Difference (APD) — falls inside a Region of Practical Equivalence (ROPE), we can accumulate positive evidence for a practically null effect rather than merely failing to reject. The APD is the natural estimand for an implementation science audience: it answers directly how many fewer households would screen positive for a mental health disorder if the intervention were delivered. Second, the Bayesian hierarchical model yields a full posterior distribution for the between-cluster variance, providing a meaningful characterization of cluster-level heterogeneity that neither GEE nor frequentist mixed-effects models directly supply.

Research Question

The Andilaye trial was powered to detect a 33% relative reduction in mental health caseness between arms (Delea et al., 2019). At the observed endline prevalences — depression 15.6%, anxiety 23.5%, emotional distress 18.5% — a 33% relative reduction corresponds to absolute reductions of approximately 5 to 8 percentage points across the three outcomes. Effects smaller in absolute magnitude than this minimum detectable difference are, by the trial's own design, below the threshold of what the study was built to find, and therefore constitute a region of practical equivalence around the null. We define the ROPE as $[-0.05, +0.05]$ on the probability-

difference scale, that is, effects smaller than 5 percentage points in either direction. This boundary is conservative relative to the trial's minimum detectable effect and directly interpretable in implementation science terms: an intervention that shifts caseness by fewer than 5 percentage points does not represent a programmatically meaningful impact regardless of its statistical precision.

Our research question is therefore: what is the posterior probability that the Average Predictive Difference (APD) — the average difference in predicted caseness probability between intervention and control households — falls within $[-0.05, +0.05]$ for each of the three mental health outcomes? And what is the full posterior distribution of between-cluster variance in mental health outcomes across the 50 kebeles?

A posterior probability above 95% inside the ROPE will be interpreted as strong positive evidence for a practically null effect. Values between 50% and 95% indicate the data are more consistent with no meaningful effect than with a practically important one, but evidence remains inconclusive. Values below 50% indicate the posterior is concentrated outside the ROPE, suggesting a practically meaningful effect cannot be ruled out.

Variables

Outcomes. Three binary indicators of household-level mental health caseness are modeled as primary outcomes. The underlying instrument is an adaptation of the Hopkins Symptom Checklist (HSCL), using 10 anxiety items (F_1_1 through F_1_10) and 13 depression items (F_2_1 through F_2_13), each scored 0 to 3. Items coded 888 or 999 were recoded as missing prior to scoring. Individual-level subscale means were computed across available items, then averaged across all eligible household members to produce one household-level score per subscale. The combined distress mean was computed across all 23 items. Binary caseness was assigned using the standard HSCL caseness threshold, which corresponds to a mean of 0.75 on the 0-to-3 item scale used here (equivalent to the published cutoff of 1.75 on the original 1-to-4 HSCL scale; see Freeman et al., 2022). The three binary outcomes are:

- **Depression** (`depression`): household mean depression score ≥ 0.75
- **Anxiety** (`anxiety`): household mean anxiety score ≥ 0.75
- **Emotional distress** (`emotion`): household mean distress score ≥ 0.75 across all 23 items

Primary predictor. Treatment assignment is a binary indicator (1 = intervention, 0 = control) assigned at the kebele level and constant within cluster. Randomization was stratified by woreda.

Fixed-effect covariate. Woreda (administrative district) is included as a fixed covariate with three levels — Farta (reference), Fogera, and Bahir Dar Zuria — represented in the model as an integer index passed directly to JAGS. This mirrors the original analysis and accounts for the stratified randomization design. Rather than dummy-coding woreda into two binary columns, a single integer index ($w_i \in \{1, 2, 3\}$) is used with a reference constraint ($\gamma_1 = 0$) imposed inside the model string, which keeps the JAGS code compact and the prior specification unified across woreda levels.

Clustering variable. Kebele is the unit of randomization and the grouping variable for random intercepts. There are 50 kebeles (25 intervention, 25 control) across the three woredas. The raw kebele identifier contains gaps and is re-indexed to a contiguous integer 1 to 50 for use in JAGS.

Data Collection and Accessibility

Data were collected through structured household surveys administered at baseline (2017) and endline (approximately 2019). The dataset is publicly accessible through the Open Science Framework (Freeman et al., 2022) and was previously obtained by the first author for a prior course project. This reanalysis uses the endline data as the primary analytical sample, consistent with the original paper's primary analysis. A panel dataset linking endline outcomes to baseline outcomes by household ID is used for a sensitivity model comparison (see Stage 2).

Analytic Sample

After applying the original study's completion filters (interview status code Z_1 = 5, indicating a completed interview), the endline sample comprises **1,472 households** nested within **50 clusters**. Cluster sizes range from 25 to 33 households per kebele (median 29). Treatment groups are exactly balanced at the cluster level: 25 intervention clusters and 25 control clusters. Woreda distribution is: Farta n = 655, Bahir Dar Zuria n = 478, Fogera n = 339.

Item-level missingness on the mental health scales is negligible: only 12 out of 4,371 individual respondents (0.3%) had any missing item on the anxiety or depression subscales. After averaging to the household level, no household in the endline sample has a missing outcome value. Complete-case analysis is therefore equivalent to full-sample analysis for all three outcomes.

Outcome prevalences in the endline sample are: anxiety 23.5%, depression 15.6%, and emotional distress 18.5%. These rates are consistent with prior HSCL-based prevalence estimates in comparable low- and middle-income country rural populations and align with those reported in Freeman et al. (2022).

Table 1. Endline analytic sample characteristics.

Characteristic	Intervention (n = 743)	Control (n = 729)
Woreda: Farta	331 (44.5%)	324 (44.4%)
Woreda: Fogera	171 (23.0%)	168 (23.0%)
Woreda: Bahir Dar Zuria	241 (32.4%)	237 (32.5%)
Households per cluster: median (range)	29 (25-33)	29 (25-32)
Anxiety caseness	176 (23.7%)	170 (23.3%)
Depression caseness	116 (15.6%)	114 (15.6%)
Emotional distress caseness	140 (18.8%)	133 (18.2%)

Note: Caseness threshold is household mean item score ≥ 0.75 on the 0-3 item scale.

Stage 2: Model Specification

Model Structure

We fit a Bayesian hierarchical logistic regression model that mirrors the frequentist mixed-effects model used in the original reanalysis. The model is specified separately for each of the three binary outcomes, following the same structural form. Let i index households ($i = 1, \dots, N$) and $j[i]$ index the kebele (cluster) to which household i belongs ($j = 1, \dots, 50$). Let $w[i] \in \{1, 2, 3\}$ denote the woreda index for household i .

Likelihood

Each binary outcome y_i is modeled as a Bernoulli draw with household-specific caseness probability p_i :

$$y_i \mid p_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{j[i]} + \beta_{\text{trt}} \cdot x_i^{\text{trt}} + \gamma_{w[i]}$$

where x_i^{trt} is the binary treatment indicator (1 = intervention, 0 = control), $\alpha_{j[i]}$ is the random intercept for kebele $j[i]$, and $\gamma_{w[i]}$ is the fixed effect for woreda $w[i]$. Farta is the reference category, enforced by the constraint $\gamma_1 = 0$.

Cluster-Level Random Intercepts

Kebele-specific random intercepts α_j are modeled as exchangeable draws from a common Normal distribution:

$$\alpha_j \mid \mu_0, \sigma_{\text{cluster}} \sim \mathcal{N}(\mu_0, \sigma_{\text{cluster}}^2), \quad j = 1, \dots, 50$$

This partial-pooling structure allows each cluster to have its own baseline log-odds of caseness while borrowing information across clusters to regularize estimates for clusters with smaller samples. Exchangeability of the kebele intercepts is justified by the study design: kebeles were randomly selected and randomly assigned, and there is no a priori reason to expect any particular kebele to differ systematically from the others in ways not already captured by the woreda fixed effects.

Prior Distributions

$$\beta_{\text{trt}} \sim \mathcal{N}(0, 1^2)$$

$$\gamma_1 = 0 \quad (\text{reference constraint, Farta})$$

$$\gamma_2, \gamma_3 \sim \mathcal{N}(0, 1^2)$$

$$\mu_0 \sim \mathcal{N}(0, 1^2)$$

$$\sigma_{\text{cluster}} \sim \text{Half-}t(3, 0, 2.5)$$

Justification for the treatment prior. The prior $\beta_{\text{trt}} \sim \mathcal{N}(0, 1)$ is weakly informative and centered at zero, reflecting genuine equipoise about the direction of the effect. On the log-OR scale, this prior places 95% of its mass between approximately -2 and +2, corresponding to odds ratios between 0.14 and 7.4. The centering at zero is justified by two bodies of evidence. First, the existing literature on WASH intervention effects on mental health consists almost entirely of observational studies showing strong cross-sectional associations between water insecurity or poor sanitation and common mental disorders — a meta-analysis of five such studies reported an odds ratio of 5.03 (95% CI: 2.26, 11.18) for the association between inadequate sanitation and common mental disorder symptoms (Ngong et al., 2023). Observational associations of this magnitude do not translate into intervention effects of similar size: the pathway from improving WASH access to reducing psychological distress involves multiple mediating steps, all of which must operate successfully at scale. Second, the Andilaye trial is one of very few cluster-randomized trials to have measured mental health as a primary outcome in this context, and its published results show prevalence ratios close to one for all three outcomes (Freeman et al., 2022). The standard deviation of 1 provides moderate regularization without strongly constraining the posterior, and is appropriate for a demand-side behavioral intervention delivered through an existing government program where large effects are a priori unlikely, particularly given documented limitations in intervention fidelity (Freeman et al., 2022).

Justification for the woreda priors. Woreda effects γ_2 and γ_3 each receive $\mathcal{N}(0, 1)$ priors, consistent in scale with the treatment prior. This unified prior specification is one advantage of the integer-index parameterization over dummy coding: a single prior statement covers both non-reference woreda effects. On the logit scale, these priors allow substantial between-woreda differences in baseline caseness rates while remaining weakly informative.

Justification for the cluster variance prior. The Half- $t(3, 0, 2.5)$ prior on σ_{cluster} is the weakly informative recommendation of Gelman (2006) for hierarchical variance parameters in logistic models. It places most prior mass on moderate between-cluster heterogeneity while allowing for the possibility of large cluster effects. In the context of the Andilaye trial, the intraclass correlation (ICC) for mental health outcomes in similar African community-based studies has been estimated between 0.01 and 0.10 (Delea et al., 2019), which corresponds to σ_{cluster} values well within the support of this prior.

Sensitivity analysis. We conduct a sensitivity analysis replacing $\beta_{\text{trt}} \sim \mathcal{N}(0, 1)$ with a more diffuse $\mathcal{N}(0, 2.5^2)$ prior to assess whether posterior inference on the treatment effect depends on the degree of prior regularization.

Primary Estimand: the Average Predictive Difference

The structural parameter β_{trt} operates on the log-odds scale — a latent, link-function artifact without direct interpretive meaning for implementation science. It is a necessary component of the model but is not itself the answer to the research question. The quantity that directly answers the research question is the **Average Predictive Difference (APD)**: the average difference in predicted caseness probability between intervention and control, computed across all households in the analytic sample while integrating over the full posterior uncertainty in all model parameters including the cluster random effects.

For each posterior draw s , the APD is computed as:

$$\widehat{\text{APD}}^{(s)} = \frac{1}{N} \sum_{i=1}^N \left[\text{logit}^{-1} \left(\alpha_{j[i]}^{(s)} + \beta_{\text{trt}}^{(s)} + \gamma_{w[i]}^{(s)} \right) - \text{logit}^{-1} \left(\alpha_{j[i]}^{(s)} + \gamma_{w[i]}^{(s)} \right) \right]$$

That is, for each household we compute its predicted probability of caseness under intervention ($x_i^{\text{trt}} = 1$) and under control ($x_i^{\text{trt}} = 0$), holding the cluster intercept and woreda effect at their posterior draw values, then average the individual differences over all $N = 1,472$ households. Repeating this across all posterior draws produces a full posterior distribution for the APD, from which we report the posterior mean, 95% Highest Density Interval (HDI), and the posterior probability that the APD falls inside the ROPE $[-0.05, +0.05]$.

The APD is a marginal, population-average effect on the natural probability scale, interpretable without reference to the logit link. A posterior mean APD of -0.03, for example, means the intervention is estimated to reduce the probability of caseness by 3 percentage points on average across the population of households in the trial. The reciprocal $1/|\widehat{\text{APD}}|$ gives the Number Needed to Treat (NNT) — the number of households that would need to receive the intervention to prevent one additional case — which we report as a secondary summary.

We also report the full posterior distribution of σ_{cluster} and the implied posterior ICC, computed draw-by-draw as:

$$\text{ICC}^{(s)} = \frac{\sigma_{\text{cluster}}^{(s)2}}{\sigma_{\text{cluster}}^{(s)2} + \pi^2/3}$$

using the logistic distribution variance $\pi^2/3$ as the within-cluster component. This quantifies the degree of residual between-kebele heterogeneity in mental health outcomes after accounting for treatment and woreda.

ROPE Specification

The ROPE is defined as $[-0.05, +0.05]$ on the APD scale, that is, a difference of fewer than 5 percentage points in absolute caseness probability between intervention and control. This boundary is anchored to the trial's own power calculation: Delea et al. (2019) powered the study to detect a 33% relative reduction in mental health outcomes. At the observed endline prevalences, a 33% relative reduction corresponds to absolute differences of 5.1 percentage points for depression, 7.8 percentage points for anxiety, and 6.1 percentage points for emotional distress. The ROPE boundary of 5 percentage points is therefore slightly conservative relative to the minimum detectable effect for depression — the most precisely estimated outcome — and directly interpretable without reference to any statistical convention: an intervention that changes caseness rates by fewer than 5 percentage points in either direction does not represent a programmatically meaningful impact in this setting.

The ROPE is applied to the full posterior distribution of the APD. The posterior probability inside the ROPE is computed as the proportion of posterior draws for which $|\widehat{\text{APD}}^{(s)}| < 0.05$. We also report the 95% HDI relative to the ROPE boundaries following Kruschke (2018): if the entire HDI falls inside the ROPE, we declare the effect practically equivalent to null; if the HDI excludes the ROPE entirely, we declare a practically meaningful effect; if the HDI overlaps the ROPE boundary, inference is inconclusive.

Model Comparison

In addition to the primary model, we fit a sensitivity model that augments the linear predictor with each household's baseline outcome as a covariate (ANCOVA adjustment):

$$\text{logit}(p_i) = \alpha_{j[i]} + \beta_{\text{trt}} \cdot x_i^{\text{trt}} + \gamma_{w[i]} + \delta \cdot y_i^{\text{bl}}$$

where $y_i^{\text{bl}} \in \{0, 1\}$ is the household's binary caseness status at baseline, and $\delta \sim \mathcal{N}(0, 1)$ is its coefficient. Including the baseline outcome is causally clean — it is measured before randomization and cannot be affected by treatment — and justified on efficiency grounds: households with higher baseline mental health burden tend to have higher endline burden regardless of treatment, and conditioning on this explains residual variance and tightens the posterior for the APD.

We compare the two models — primary (woreda + cluster) and sensitivity (woreda + cluster + baseline outcome) — using Pareto-smoothed importance sampling leave-one-out cross-validation (PSIS-LOO; Vehtari et al., 2017). If LOO favors the ANCOVA model by more than 2 standard errors in expected log predictive density (ELPD), we report both sets of APD estimates and discuss the discrepancy. If the models are indistinguishable by LOO, the primary model is preferred on grounds of parsimony and because the 4 households with missing baseline outcomes are retained without any additional assumption.

Of the 1,472 endline households, 1,468 have a baseline record; the 4 remaining households had incomplete baseline interviews ($Z_1 \neq 5$ at baseline) and have missing values for y_i^{bl} . These 4 households are excluded from the ANCOVA model but retained in the primary model. LOO is

computed on the common sample of 1,468 households to ensure the comparison is on identical data.

Stage 3: Prior Predictive Check

Simulation Procedure

Prior predictive simulation draws parameters from the prior distributions alone — without conditioning on any observed outcome data — and uses those draws to generate hypothetical datasets. This reveals what kinds of data the model considers plausible before seeing any evidence, and verifies that the prior specification is neither too restrictive (concentrating all probability mass on an implausibly narrow range of outcomes) nor too diffuse (assigning meaningful probability to outcomes that are physically or substantively impossible).

We simulate 1,000 prior predictive datasets using the following procedure. For each draw $s = 1, \dots, 1000$:

1. Draw $\mu_0^{(s)} \sim \mathcal{N}(0, 1)$
2. Draw $\sigma_{\text{cluster}}^{(s)} \sim \text{Half-}t(3, 0, 2.5)$
3. Draw $\alpha_j^{(s)} \sim \mathcal{N}(\mu_0^{(s)}, \sigma_{\text{cluster}}^{(s)2})$ for each of the 50 clusters
4. Draw $\beta_{\text{trt}}^{(s)} \sim \mathcal{N}(0, 1)$
5. Set $\gamma_1^{(s)} = 0$; draw $\gamma_2^{(s)}, \gamma_3^{(s)} \sim \mathcal{N}(0, 1)$
6. Compute $\text{logit}(p_i^{(s)}) = \alpha_{j[i]}^{(s)} + \beta_{\text{trt}}^{(s)} \cdot x_i^{\text{trt}} + \gamma_{w[i]}^{(s)}$ for all $N = 1,472$ households, using their observed treatment and word index values
7. Draw $y_i^{(s)} \sim \text{Bernoulli}(p_i^{(s)})$
8. Compute the implied sample-level caseness prevalence $\bar{y}^{(s)} = \frac{1}{N} \sum_i y_i^{(s)}$
9. Compute the implied prior predictive APD: $\widehat{\text{APD}}^{(s)}$ as defined in Stage 2, applied to the prior draw

The R code for this simulation is included in the reproducibility bundle

`(01_prior_predictive.R)`.

Results

Prevalence distribution. The prior predictive distribution for sample-level caseness prevalence is shown in Figure 1. Under the chosen priors, the distribution of predicted prevalences spans the full range from near 0% to near 100%, with a median near 50% and substantial mass between 5% and 95%. This confirms the priors are weakly informative: the model does not pre-commit to any particular prevalence regime. The observed endline prevalences (depression 15.6%, anxiety 23.5%, distress 18.5%) lie comfortably within the prior predictive distribution, confirming the prior does not conflict with the data before any conditioning has occurred.

Prior APD distribution and ROPE. Figure 2 shows the prior predictive distribution of the APD mapped onto the ROPE boundaries of $[-0.05, +0.05]$. Because the prior on β_{trt} is centered at zero and the APD is a nonlinear transformation of the log-OR through the logistic function, the prior predictive APD is also centered near zero but has heavier tails than the log-OR prior, reflecting the nonlinearity of the inverse-logit transformation. The prior assigns substantial probability both inside and outside the ROPE, confirming the ROPE analysis is not predetermined: posterior concentration inside or outside the ROPE will be driven by the data rather than the prior specification.

Cluster heterogeneity. The prior on σ_{cluster} generates cluster-level baseline caseness probabilities that range widely across simulated draws, allowing the model to accommodate both homogeneous clusters and highly heterogeneous ones. The implied prior ICC — computed draw-by-draw as $\sigma_{\text{cluster}}^{(s)2} / (\sigma_{\text{cluster}}^{(s)2} + \pi^2/3)$ — has a median of approximately 0.20 with a wide 95% credible interval, which is appropriate given the limited prior evidence on clustering of mental health outcomes specifically in kebele-level Ethiopian community trials.

Overall, the prior predictive check confirms the chosen priors are appropriately weakly informative: they are consistent with a broad range of plausible outcomes for a community-based intervention trial in this setting, without generating pathological predictions such as prevalences of exactly 0 or 100%, or ICC values inconsistent with cluster-randomized trial designs.

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